AI ACADEMY · LEVEL 3 · GRADE 8 · WEEK 10 · STUDENT LESSON

Image Generation Models

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

Before you start

Week 1: generated content is not an observation of an event.

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Explain the mechanism and terminology of image generation and provenance. Try the recall before opening notes.

Your learning target

I can explain the **mechanism** and terminology of image generation and **provenance**.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Prompt Evaluation. Give an example and a mistake that the idea helps you avoid.

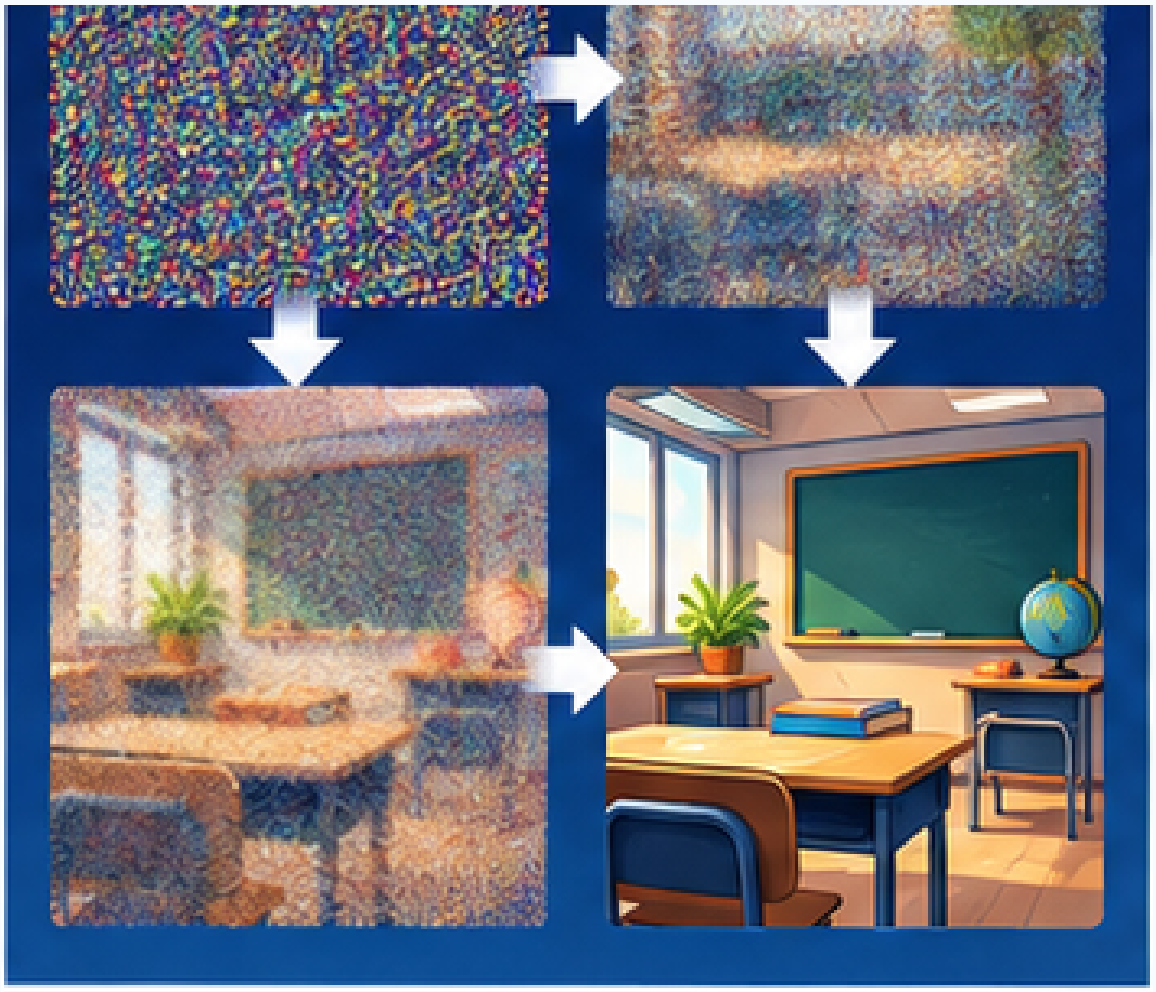
Without opening the old lesson, explain one key idea from Iterative Prompting. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Sampling and Variability. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Week 10: Image Generation Models

Image generation iteratively constructs pixels guided by learned representations.



CREATOR MISSION

Sequence a simplified **noise**-to-image card model.

Creator name and date:

> **Write or draw your reasoning and evidence.**

Readiness for the connected explanation

Recall Generation failures and verification. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

This week's end goal

By the end, I can describe a toy **denoising process** and distinguish generated illustrations from **observation**s.

STUDENT LESSON · 2 OF 18

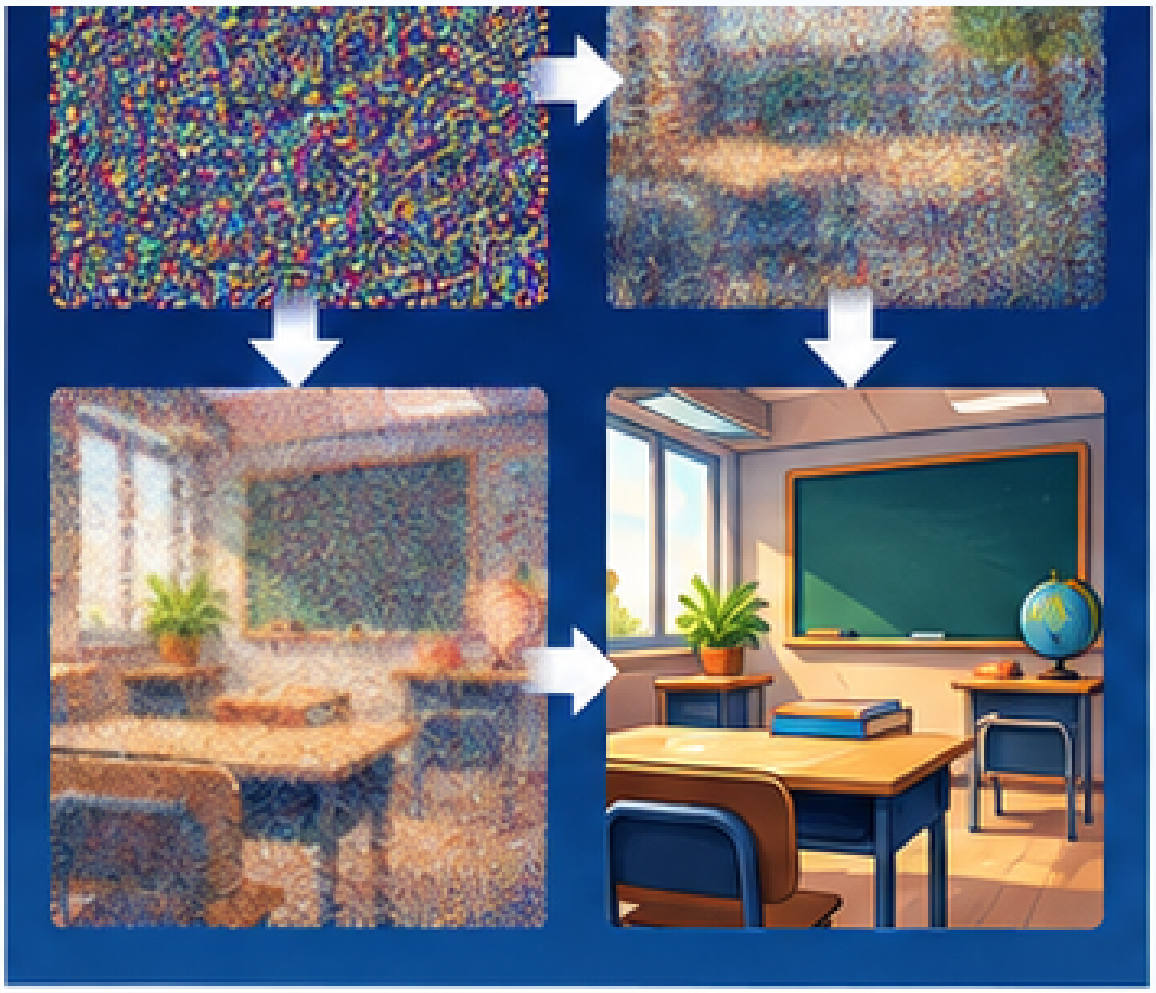
Notice the evidence

> Investigate the supplied Image Generation Models case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE STORY

Notice → Predict → Explain

Begin with visible **evidence**. Do not explain more than the picture and information support.



Start with this puzzle

MYSTERY

The learners must use image generation models to solve a problem. What information do they have? What decision or calculation do they make? What would count as **evidence** that it worked?

Three **observation**s (what you can point to):

> **Write or draw your reasoning and evidence.**

My **prediction** and the clue supporting it:

> **Write or draw your reasoning and evidence.**

A closer explanation

An image generator constructs an **output** from learned patterns and a specification. A generated picture is not a record that the depicted event occurred.

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

STUDENT LESSON · 3 OF 18

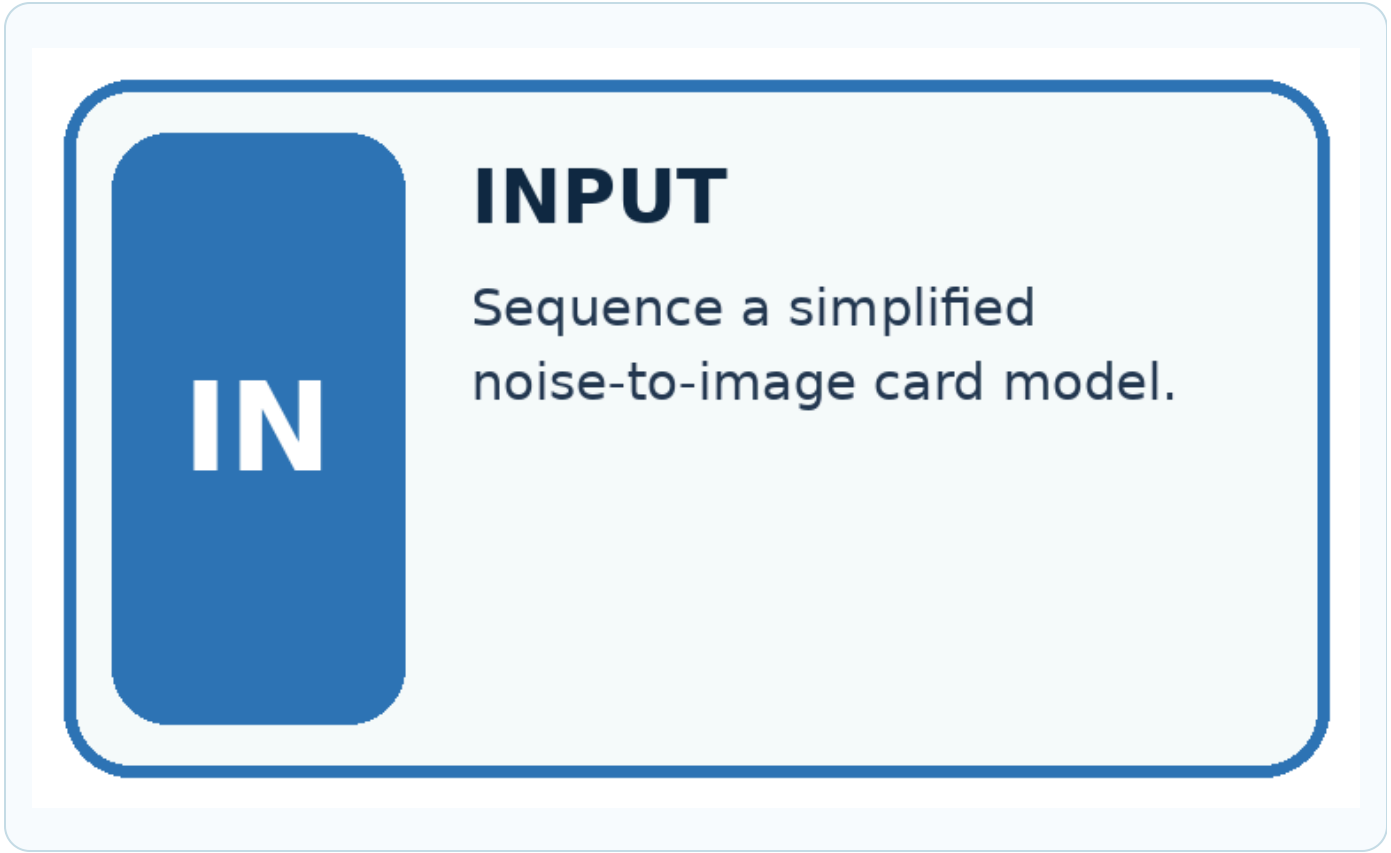
See how it works

> Follow the mechanism for Image Generation Models. Connect each labelled input, deciding step and result.

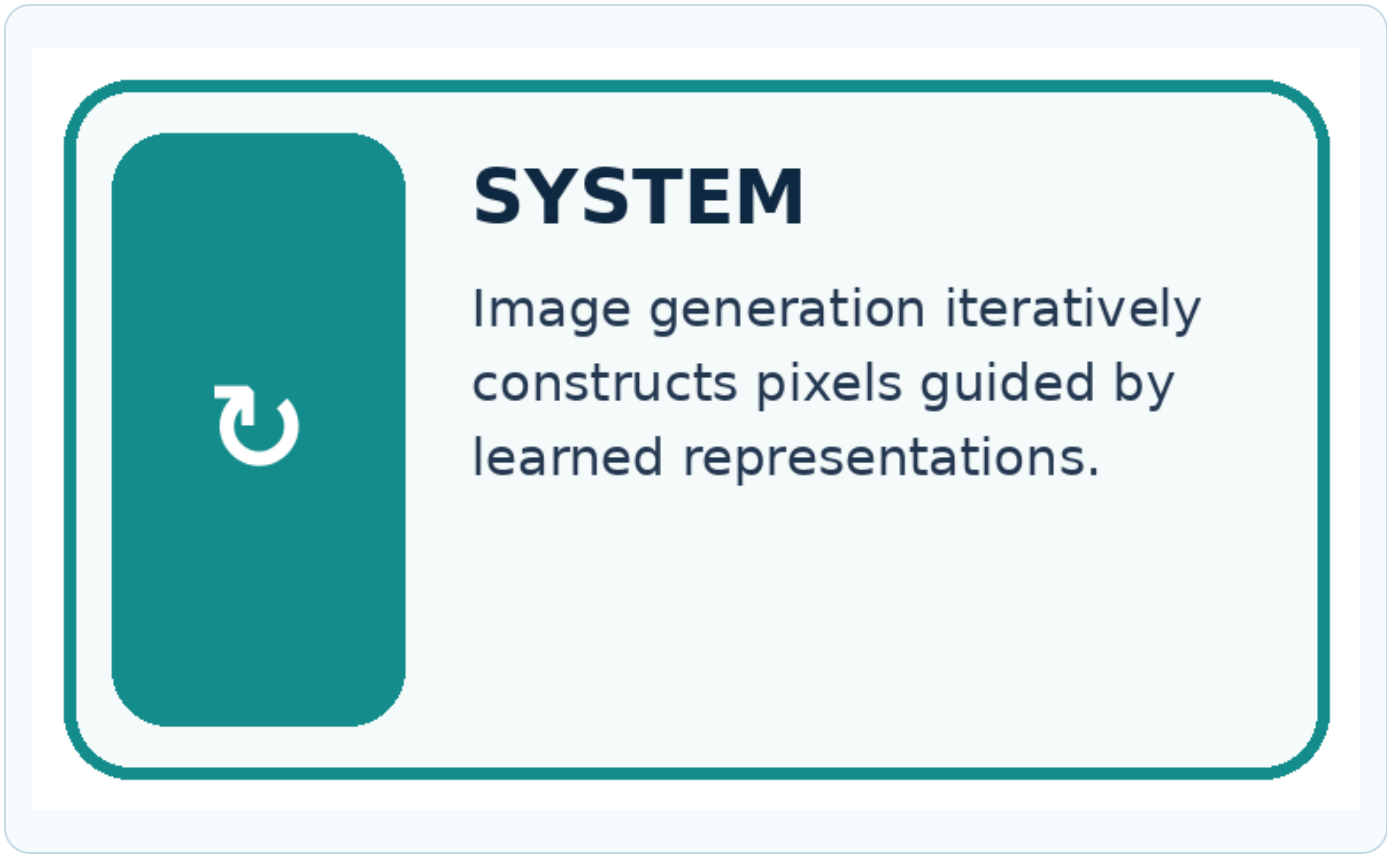
VISUAL MECHANISM

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.



↓


SYSTEM
Image generation iteratively constructs pixels guided by learned representations.
OUT
EVIDENCE
Sequence a simplified noise-to-image card model.

Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation

Keep a **provenance** record containing specification, tool/version when known, generation date, edits, permissions and intended use. Label fictional imagery.

Inspect visual requirements and avoid inferring identity or event authenticity from appearance.

Image generation and provenance — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
A fictional poster requires exactly three labelled circles. Generated image shows four circles and one unreadable label. A caption calls it a photograph of a real lesson.	Keep a provenance record containing specification, tool/version when known, generation date, edits, permissions and intended use.	It fails count and label criteria; the photograph claim is unsupported. Record it as a generated fictional illustration and revise or reject before use.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

Compare the cases

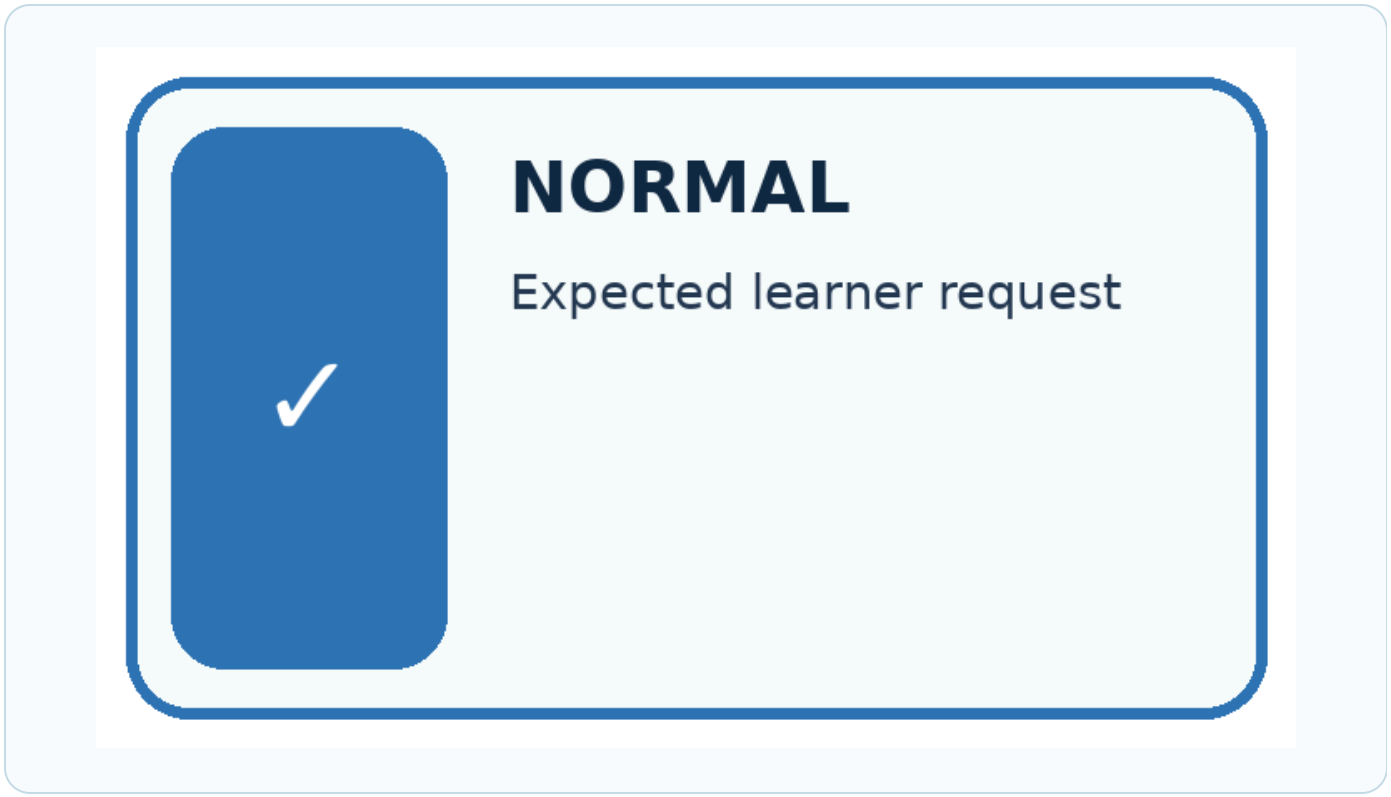
> Compare the supplied cases for Image Generation Models. Identify the detail that changes the conclusion.

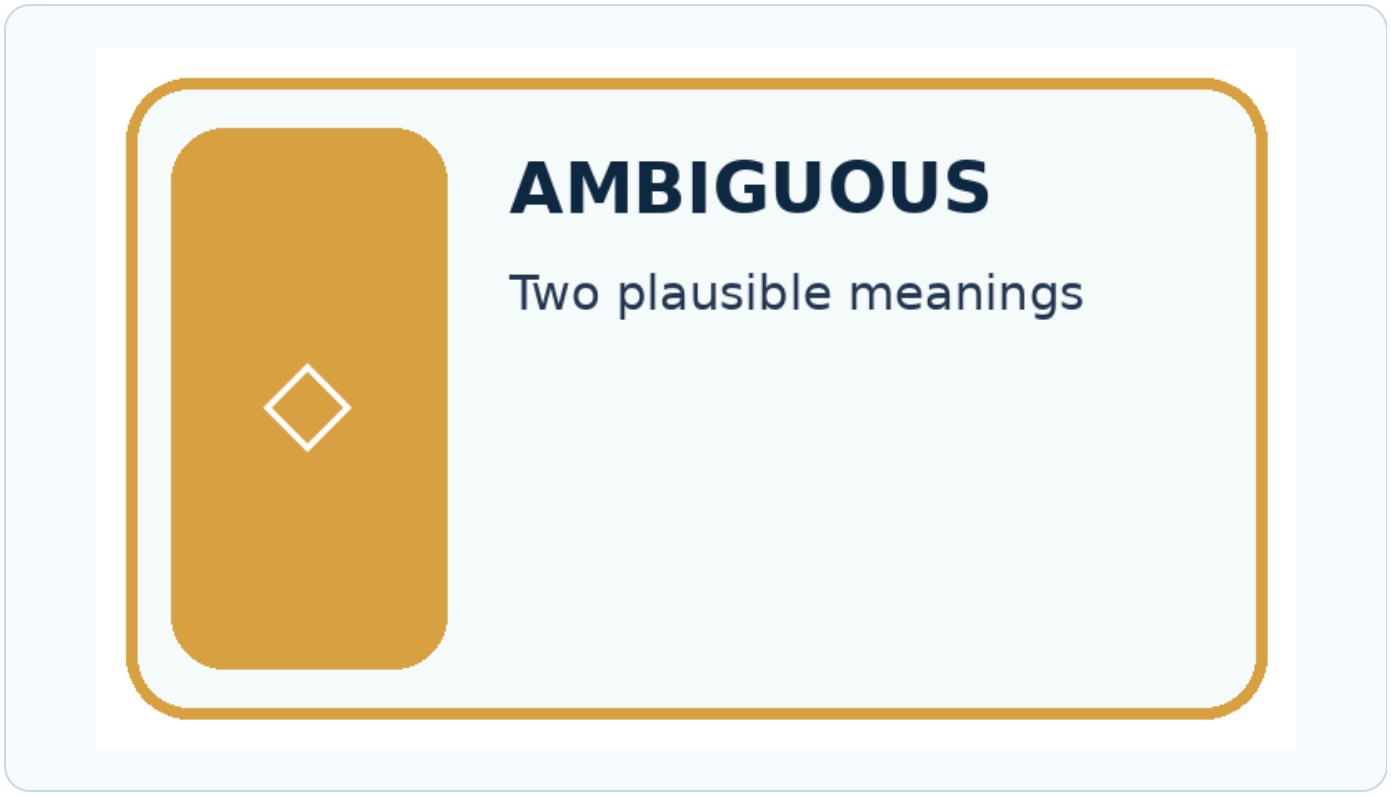
The case to investigate

CASE GALLERY

Four Tests, Four Kinds of **Evidence**

A system is not understood until it survives more than one comfortable example.


NORMAL
Expected learner request


AMBIGUOUS
Two plausible meanings

The normal case

NORMAL CASE: predict and name **evidence**. BOUNDARY CASE: predict and name **evidence**.





Check missing information

MISSING INFORMATION: predict and name **evidence**. TRANSFER CASE: predict and name **evidence**.

Which case is most likely to reveal a hidden assumption? Why?

Write or draw your reasoning and evidence.

Change one thing in the pictured case

Change one thing

Change: Relabel the illustration as a photograph of an actual laboratory measurement.

Predicted result

Prediction: That caption is unsupported.

Why the result follows

Reason: Its **provenance** is an invented diagram of a **process**, not a camera record of an event.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

> Use the vocabulary of Image Generation Models to explain a concrete decision, rather than only memorising definitions.

VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

1

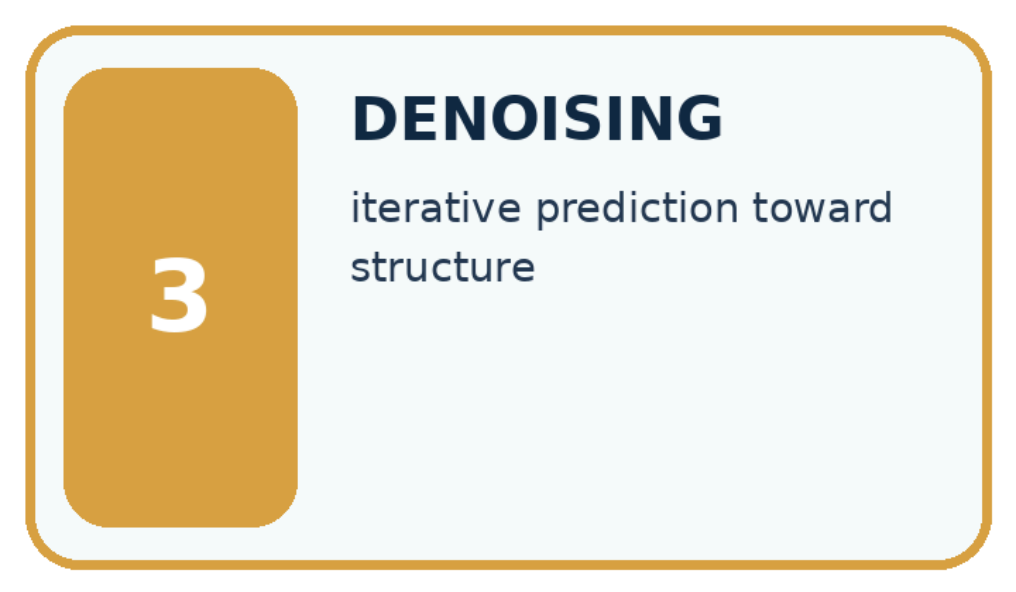
LATENT REPRESENTA

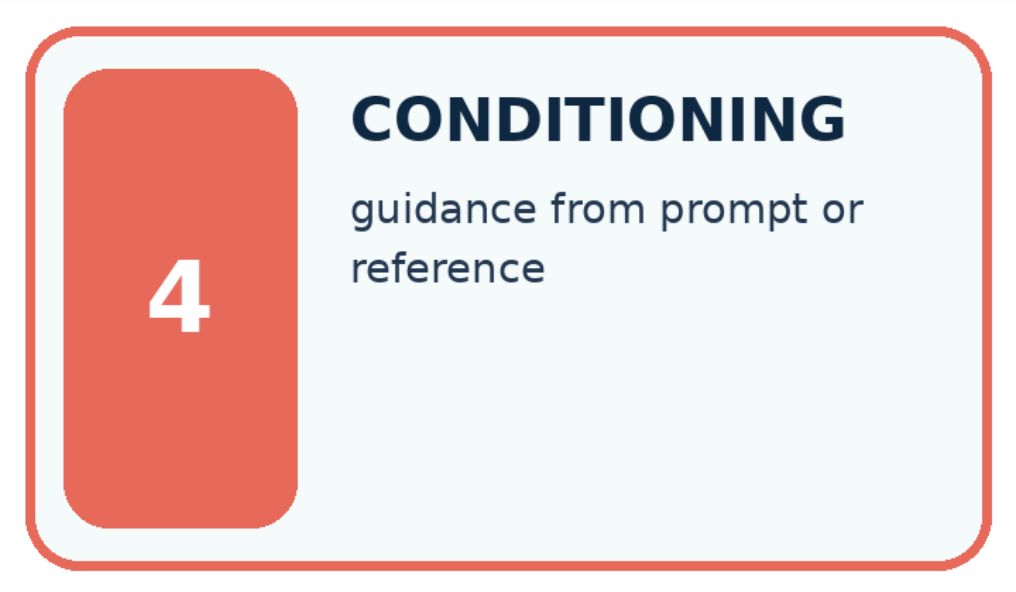
compressed learned pattern space

2

NOISE

random starting signal





Word	My own example	Not this / boundary
latent representation	An internal representation learned or constructed by a model rather than directly supplied as the original observation.	
noise	Variation or interference that obscures the information relevant to a task.	

Word	My own example	Not this / boundary
denoising	Reducing noise in a representation; some generative models learn repeated denoising steps to create outputs.	
conditioning	Making a model's output depend on supplied information, such as a prompt or image.	

Words used in the closer explanation

Word	Meaning
Provenance	Recorded origin and transformation history
Visual criterion	A stated requirement checked against an image
Authenticity claim	A claim about the source or reality of depicted content

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STUDENT LESSON · 6 OF 18

Follow a worked example

> Follow the worked trace for Image Generation Models, predict a step, then compare with its explanation.

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

Worked evidence for Image Generation Models

STEP	EVIDENCE AND REASONING
1	In this simplified diffusion-style image generator, begin with random visual noise. Other image-generation methods use different procedures.
2	At each step, predict a slightly more structured image guided by the prompt.
3	Learned patterns influence shapes, textures, and relationships.
4	The result is generated, not retrieved evidence of a real event.

Trace these corrected steps. Keep the stated metric, assumptions and interpretation when you change the case.

1. In this simplified diffusion-style image generator, begin with random visual **noise**. Other image-generation methods use different procedures.

2. At each step, predict a slightly more structured image guided by the prompt.

3. Learned patterns influence shapes, textures, and relationships.

4. The result is generated, not retrieved **evidence** of a real event.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

A fictional poster requires exactly three labelled circles. Generated image shows four circles and one unreadable label. A caption calls it a photograph of a real lesson.

It fails count and label criteria; the photograph claim is unsupported. Record it as a generated fictional illustration and revise or reject before use.

Worked explanation · Image Generation Models

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Image Generation Models

A fictional lesson poster illustrates one family of image-generation approaches.

The facts we will use

Supplied facts

Toy 2×2 noisy numerical grid: [[0.2,0.8],[0.6,0.3]]. A specified illustrative **denoising** step gives [[0.1,0.9],[0.4,0.2]].

Supplied facts

These supplied arrays illustrate steps only; they are not **output**s from a real trained diffusion model.

AI ACADEMY / AI-3/10

Illustrative grids, not executed model outputs

	Left	Right
Before top	0.2	0.8
Before bottom	0.6	0.3
After top	0.1	0.9
After bottom	0.4	0.2

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-3/10

Image Generation Models

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Noisy representation

Four toy numerical cells form the starting grid.



2. Condition on request

A text request guides the imagined learned denoising process.



3. Illustrative update

Use the supplied second grid to show one change in representation.



4. Check provenance

Caption it as a generated-process illustration, not a photographed experiment.

CHANGE ONE THING

Relabel the illustration as a photograph of an actual laboratory measurement.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Noisy representation	Four toy numerical cells form the starting grid.
2. Condition on request	A text request guides the imagined learned denoising process.
3. Illustrative update	Use the supplied second grid to show one change in representation.
4. Check provenance	Caption it as a generated-process illustration, not a photographed experiment.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Diffusion-style generators learn to remove **noise** in stages, often guided by text, but a real implementation can operate in a **latent representation** rather than direct pixels.

The toy values let us inspect an update without pretending to simulate that trained **process**. An attractive final image still needs checks for labels, requested relations and factual claims.

What this example does not establish

Keep the stated limit

Not all image generators use diffusion; this toy step is neither a complete algorithm nor an executed model.

Keep the stated limit

Read the labelled picture: What may these two grids explain, and what can they not establish about a real generator?

Keep the stated limit

Change one thing: Relabel the illustration as a photograph of an actual laboratory measurement. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Image Generation Models claim and explain a repair using the supplied evidence.

Find the first error

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak **evidence**.

!

COMPARE • DIAGNOSE

Weak idea: An image model searches a hidden folder and copies the closest picture. Strong move: inspect the claim, source, stage, schema, or test evidence; repair one cause; rerun.

Evidence record	Expected	Actual	Likely cause / next test
normal case	works		
boundary case	stated rule		
missing input	safe response		
fresh transfer	generalizes		

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

Check the claim against the facts

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.

STUDENT LESSON · 8 OF 18

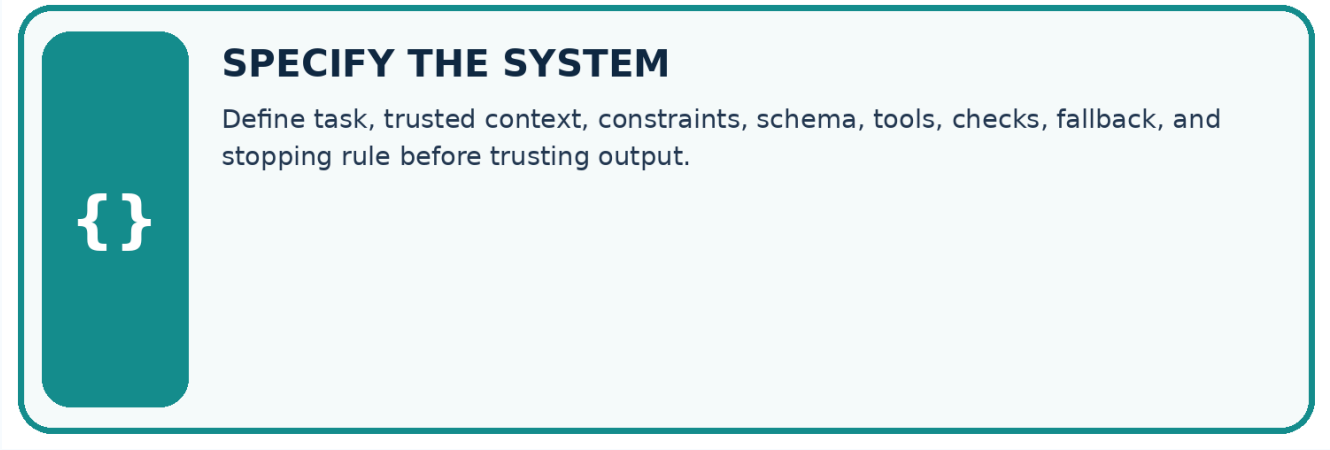
Practise together

Practise Image Generation Models with support, then explain the deciding step in your own words.

LOGIC STUDIO

Specify Before You Run

Prompts, schemas, retrieval, tool calls, and code are inspectable parts of one AI product.



SET task, audience, constraints, **output**_schema LOAD approved_context or retrieval_result CALL model or allowed_tool VALIDATE fields, claims, citations, and safety IF **evidence** missing: ask, retrieve, or abstain RETURN useful **output** + source trail + limitations

> Trace one **input**. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

Provenance tracks production history; it does not prove the content is accurate or establish every legal permission.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Practise with less help: Trace the worked case

Use workbook W2 for the connected example “Image generation and **provenance**”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Image generation and **provenance**. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

What may these two grids explain, and what can they not establish about a real generator?

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

Run the investigation

> Investigate Image Generation Models. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



Order **noise**-to-image cards, vary one prompt condition, compare composition changes, and label what cannot be inferred as fact.

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

> **Test / input:** normal
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

> **Test / input:** boundary
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

> **Test / input:** missing
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

> **Test / input:** fresh transfer
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

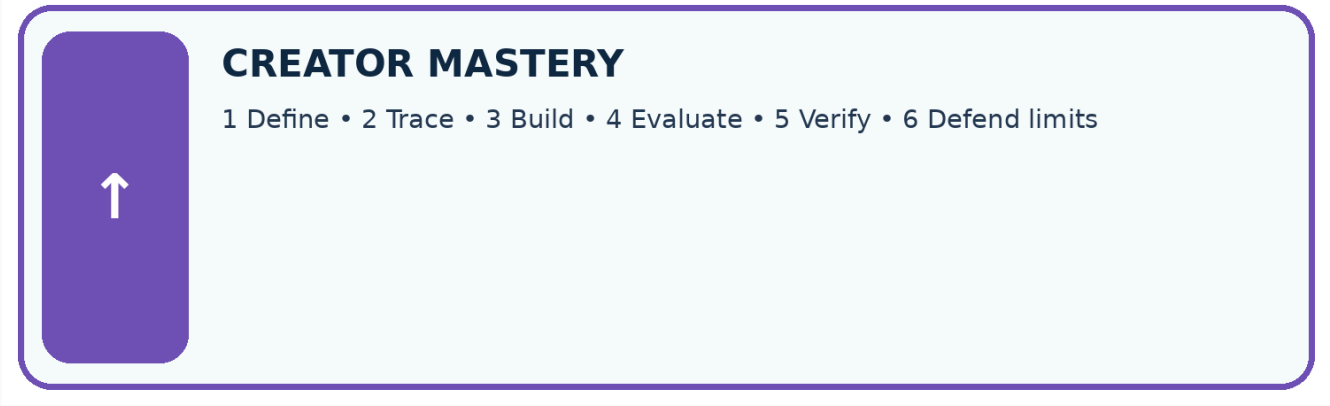
Build your understanding

> Move from recall to tracing, diagnosis and transfer in Image Generation Models. Explain why each step is justified.

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the worked example.



1 • NAME: Define the concept without repeating the textbook.

2 • TRACE: Show the information, state, branch, calculation, or **evidence** path.

3 • APPLY: Solve a different example independently.

4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.

5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

My strongest **evidence** of independent understanding:

> **Write or draw your reasoning and evidence.**

STUDENT LESSON · 11 OF 18

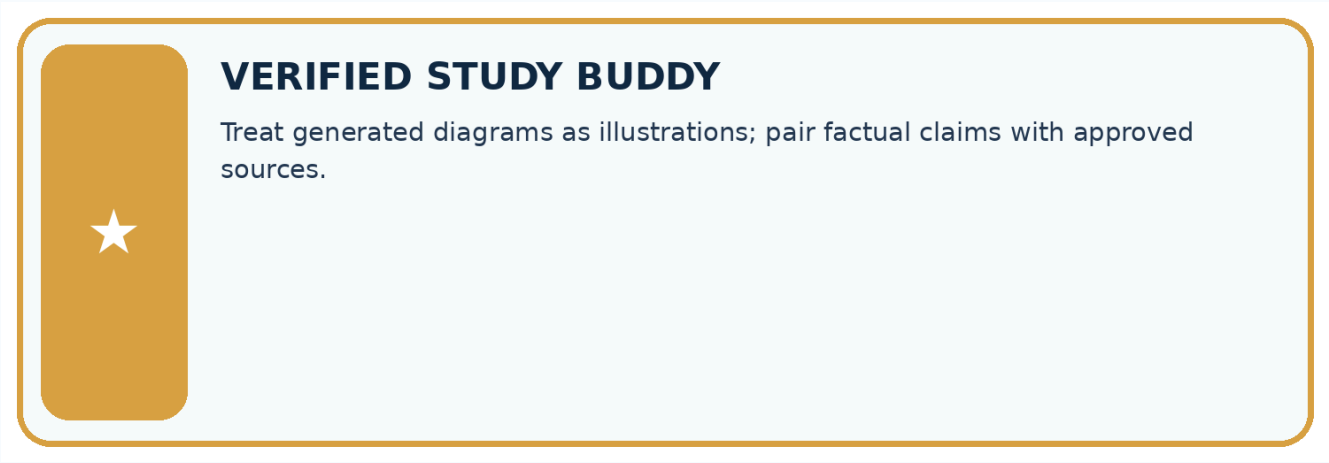
Connect your project

> Apply this week's idea to Verified Study Buddy. Keep an annotated decision and the evidence supporting it.

CUMULATIVE PROJECT

Verified Study Buddy

Every week adds one inspectable piece. The system must show how it reached a result.



THIS WEEK'S CHECKPOINT Treat generated diagrams as illustrations; pair factual claims with approved sources.

Trace the system

SYSTEM MAP Learner need → prompt/schema → approved sources/retrieval → model/tool → structured answer → verification → **human decision**

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under **human control**?

Write or draw your reasoning and evidence.

Responsible design

RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse fluent **output** with verified truth.

Evidence for your project

Use Study Buddy generated diagrams only with clear **provenance** and checked educational labels.

STUDENT LESSON · 12 OF 18

Show what you know

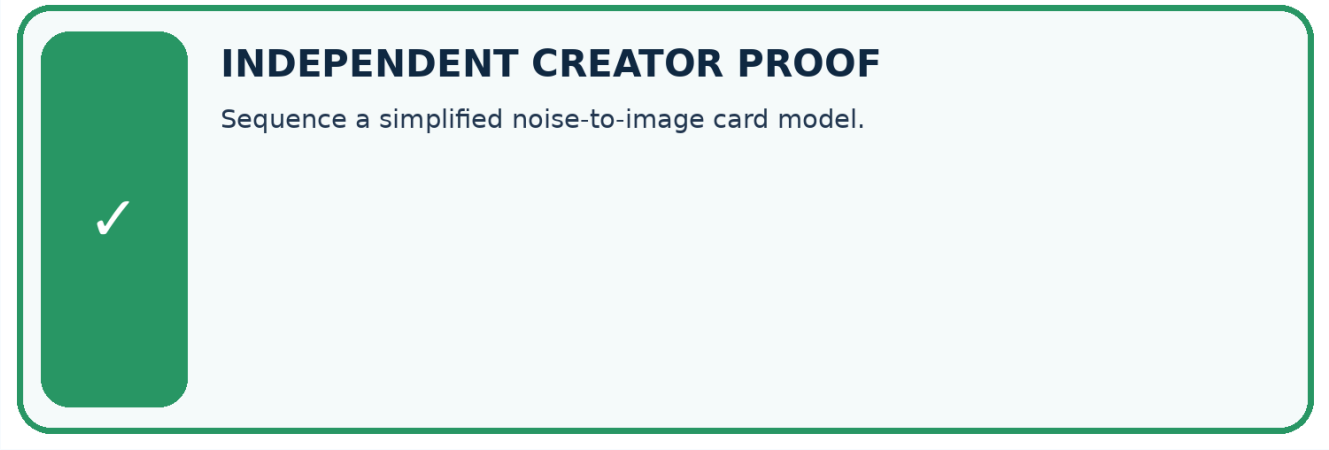
Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Independent evidence

INDEPENDENT MASTERY PROOF

Explain • Transfer • Test • Defend

Complete this page without copying the worked example. Your reasoning matters more than decoration.



Try a more demanding case

CHALLENGE Sequence a simplified **noise**-to-image card model.

1. My fresh example, trace, calculation, or design:

Write or draw your reasoning and evidence.

2. My **evidence** that it works (include a boundary or missing-data case):

Write or draw your reasoning and evidence.

3. One limitation, fairness concern, or situation needing human review:

Write or draw your reasoning and evidence.

Check your explanation

SELF-CHECK ☐ I explained the **mechanism** ☐ I used **evidence** ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

STUDENT LESSON · 13 OF 18

Study a complete case

Revisit the complete Image Generation Models case for supported practice. Repeated examples do not become fresh assessments.

Week 10 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

A generator creates a realistic fictional bicycle beside a tree.

The question

> How does this differ from event photography?

Worked explanation

It synthesises an image from learned patterns and instructions; it does not document that event. Many systems de**noise** iteratively, but architectures vary.

Explain it in your own words

> Which detail supports the answer, and why does it matter?

> **Write or draw your reasoning and evidence.**

Trace the supplied case visually

A generated image does not document an event

Evidence or stage	Value or result
Brief	Bicycle beside a tree
Model	Synthesises visual patterns
Image	Fictional scene
Check	Content and intended use

A generator creates a realistic fictional bicycle beside a tree.

> How does this differ from event photography?

It synthesises an image from learned patterns and instructions; it does not document that event. Many systems de**noise** iteratively, but architectures vary.

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Apply a new case independently

A generated diagram requires two arrows and labels **input**/**output**. It has two arrows but swapped labels. Metadata records generation, yet the caption says experimental **observation**. Audit.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 10 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

> An image model draws a plant with seven roots after a request for exactly four. What does this show, and where should an exact count diagram be made?

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

Write or draw your reasoning and evidence.

Connect to your project

Treat generated diagrams as illustrations; pair factual claims with approved sources.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Transfer practice from the original programme

An image generator produces a convincing picture of a school greenhouse on Mars. A student calls it photographic **evidence**. Explain what the image can and cannot establish.

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Image Generation Models, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: A fictional lesson poster illustrates one family of image-generation approaches.

By the end, I can	How I will show it
Explain the mechanism and terminology of image generation and provenance	Explain the deciding step, show a trace or test, and state one limit.

By the end, I can	How I will show it
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Image Generation Models without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of image generation and **provenance**

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional lesson poster illustrates one family of image-generation approaches.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Caption it as a generated-**process** illustration, not a photographed experiment.

1. Noisy representation

2. Condition on request

3. Check **provenance**

4. Illustrative update

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Check **provenance** Caption it as a generated-**process** illustration, not a photographed experiment.

Diffusion-style generators learn to remove **noise** in stages, often guided by text, but a real implementation can operate in a **latent representation** rather than direct pixels.

The toy values let us inspect an update without pretending to simulate that trained **process**. An attractive final image still needs checks for labels, requested relations and factual claims.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Relabel the illustration as a photograph of an actual laboratory measurement. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. That caption is unsupported.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: That caption is unsupported. Its **provenance** is an invented diagram of a **process**, not a camera record of an event.

Repair a missing building block

Review Prompt Evaluation (ai-3/9). Goal: Explain the **mechanism** and terminology of controlled prompt experiments. Short practice: Why is B's selection defensible here without calling it universally superior?.

Return to this week and explain what you changed.

Review Iterative Prompting (ai-3/7). Goal: Explain the **mechanism** and terminology of controlled prompt experiments. Short practice: What **evidence** supports improvement here, and what remains unproven?.

Return to this week and explain what you changed.

Review Sampling and Variability (ai-3/4). Goal: Sample from a distribution using explicit cumulative intervals. Short practice: Does T=0.5 make B impossible, and does choosing A guarantee factual accuracy?.

Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

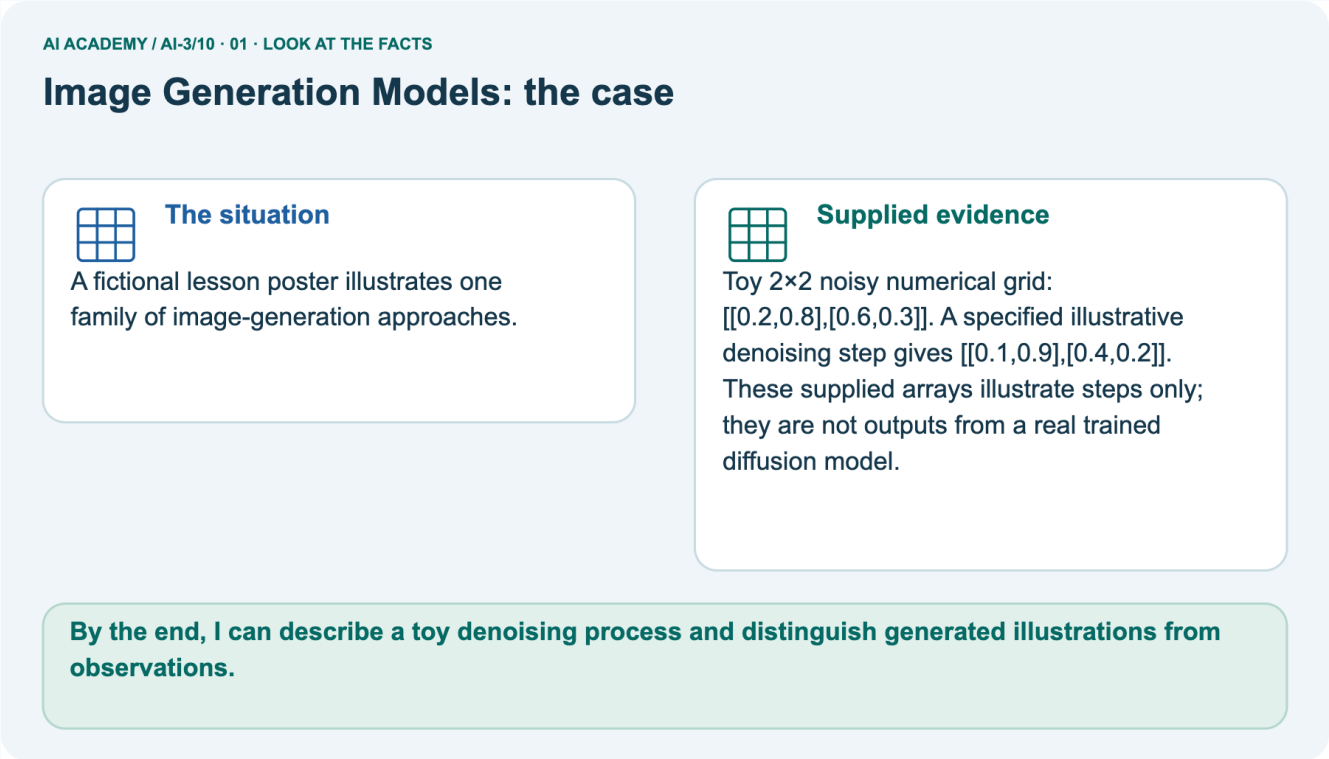
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Image Generation Models graphic using its labels, deciding evidence and relevant limitation.

01. Image Generation Models: the case



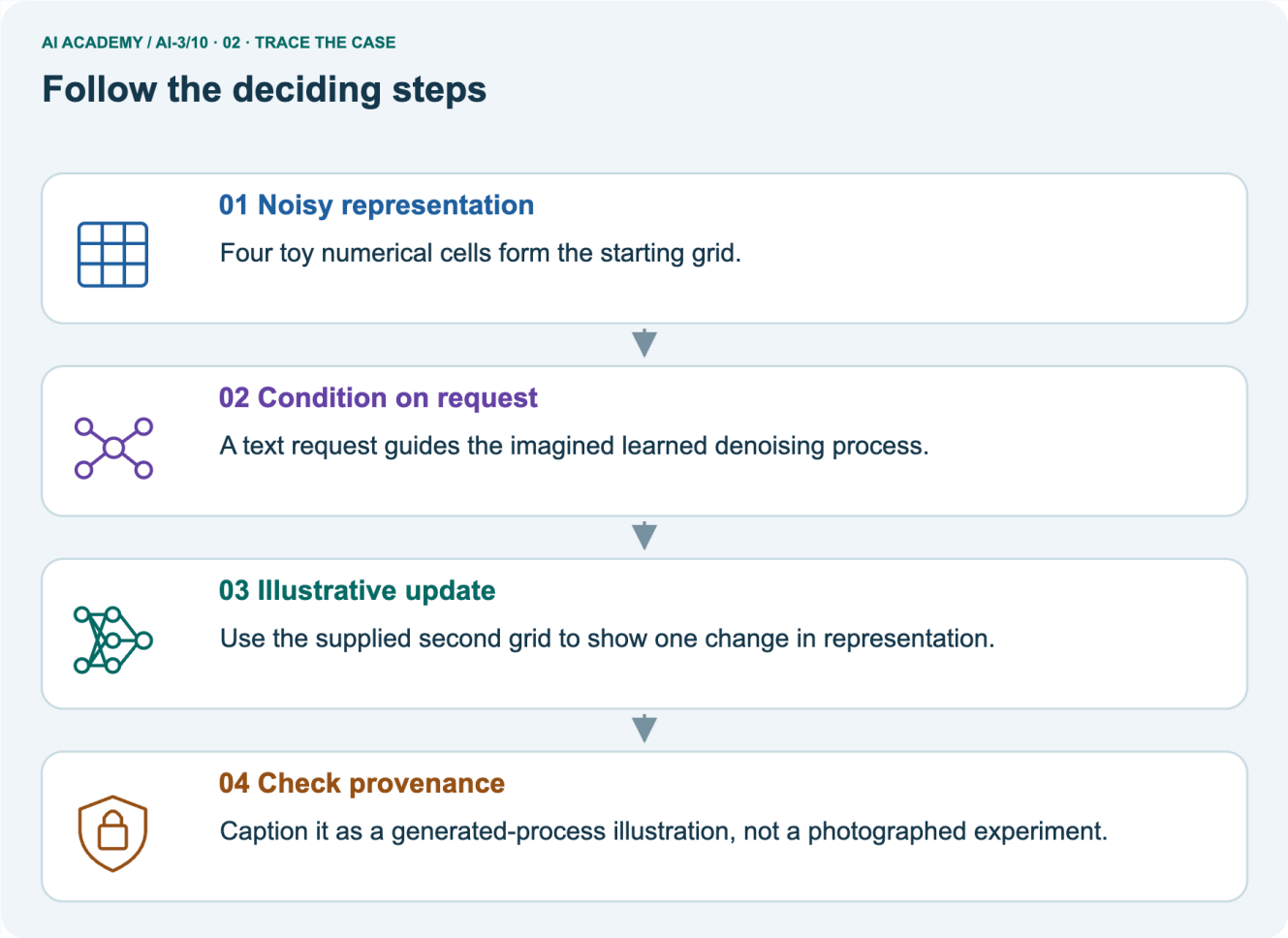
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional lesson poster illustrates one family of image-generation approaches. Supplied facts: Toy 2×2 noisy numerical grid: [[0.2,0.8],[0.6,0.3]]. A specified illustrative **denoising** step gives [[0.1,0.9],[0.4,0.2]].

These supplied arrays illustrate steps only; they are not **output**s from a real trained diffusion model.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

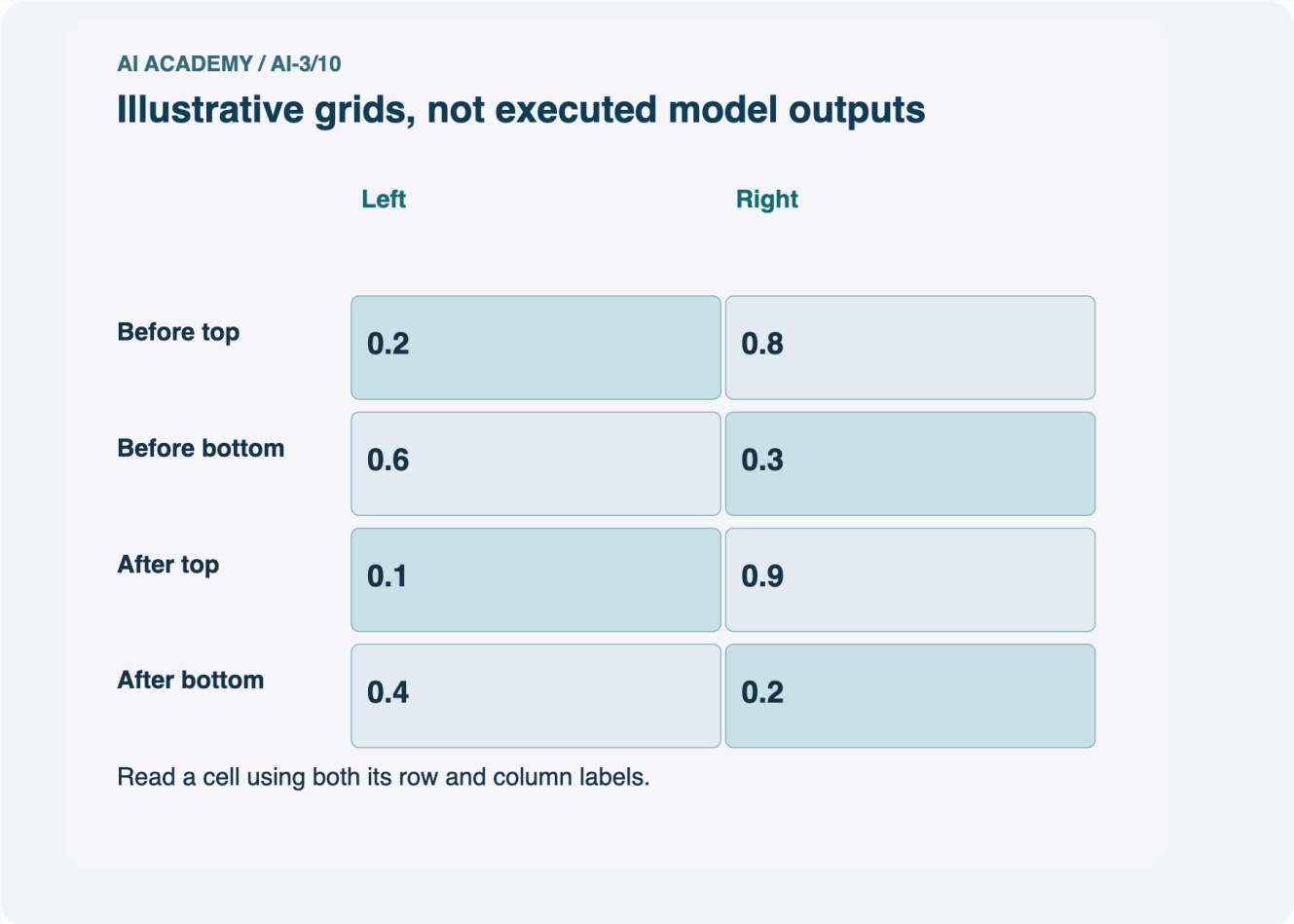


Follow the labelled picture with the matching explanation. Open the vector diagram

Noisy representation: Four toy numerical cells form the starting grid. → Condition on request: A text request guides the imagined learned **denoising process**. → Illustrative update: Use the supplied second grid to show one change in representation. → Check **provenance**: Caption it as a generated-**process** illustration, not a photographed experiment.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Illustrative grids, not executed model outputs

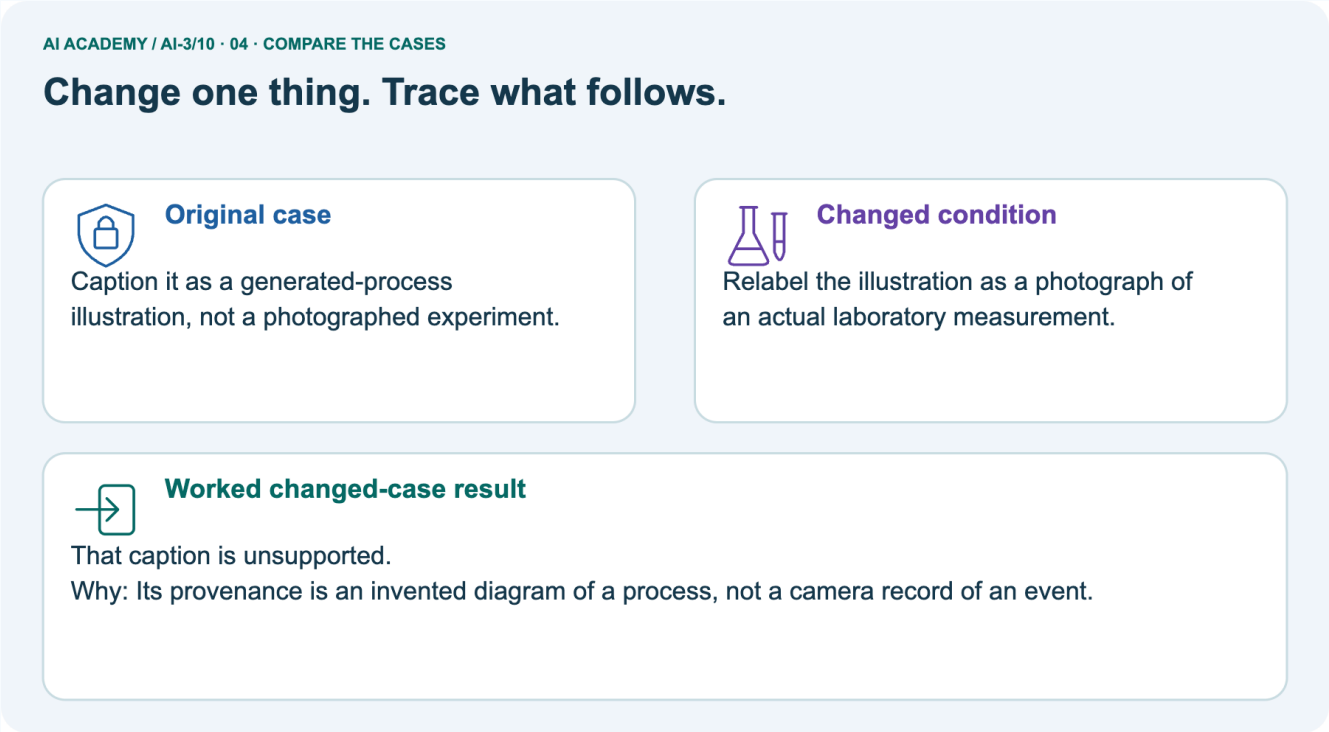


Follow the labelled picture with the matching explanation. Open the vector diagram

Illustrative grids, not executed model **output**s. Before top: Left = 0.2, Right = 0.8. Before bottom: Left = 0.6, Right = 0.3. After top: Left = 0.1, Right = 0.9. After bottom: Left = 0.4, Right = 0.2.

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



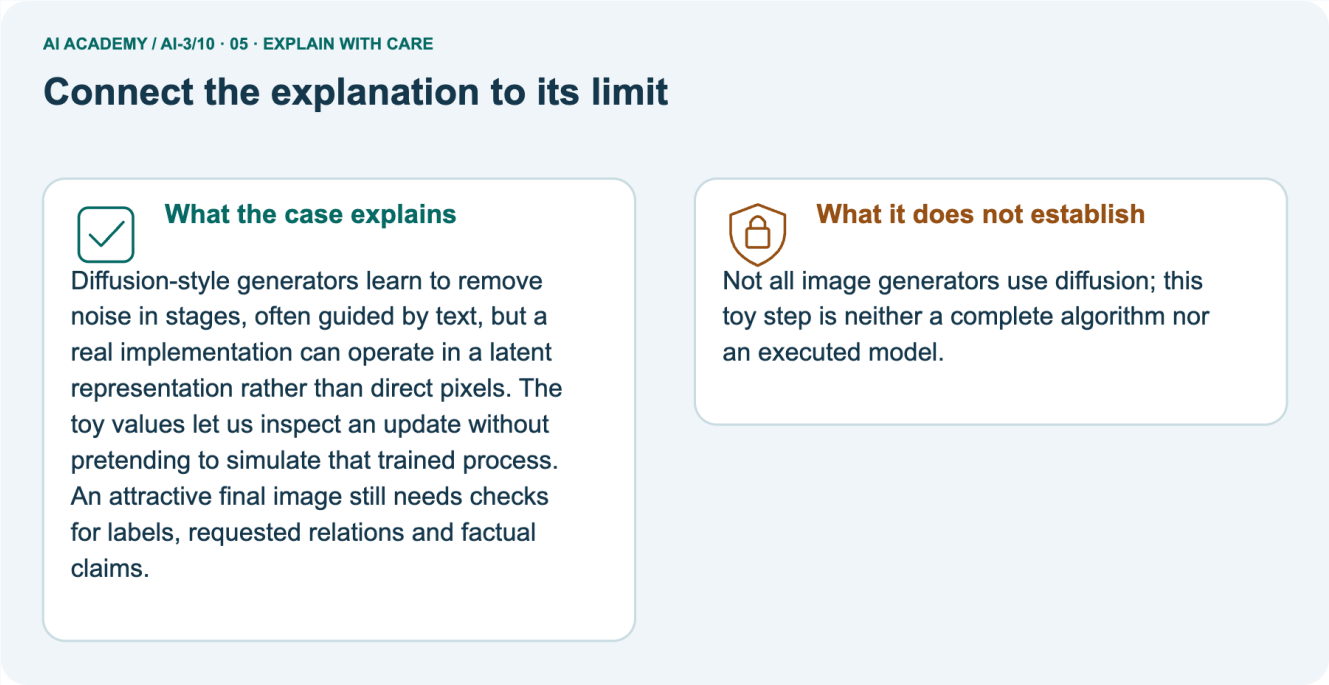
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Caption it as a generated-**process** illustration, not a photographed experiment.
Changed condition: Relabel the illustration as a photograph of an actual laboratory measurement.

Expected result: That caption is unsupported. Reason: Its **provenance** is an invented diagram of a **process**, not a camera record of an event.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

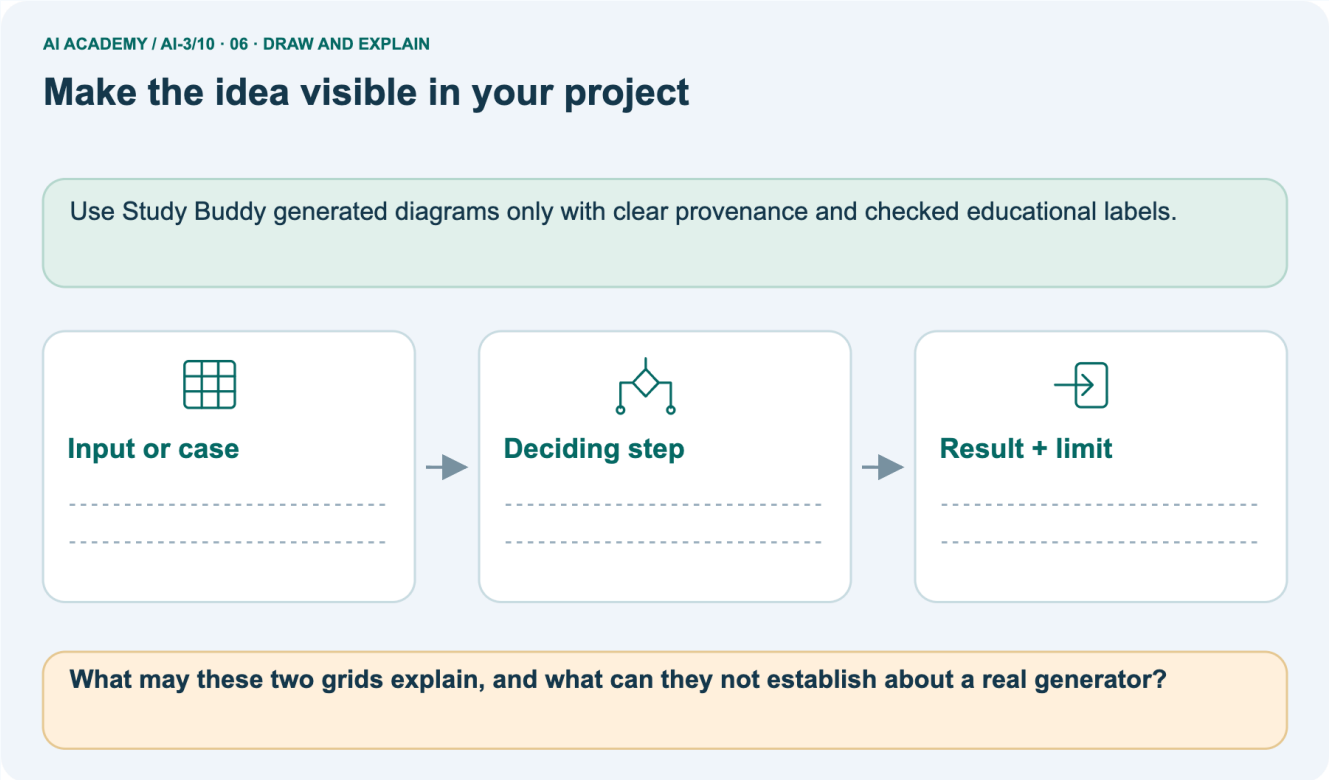
Diffusion-style generators learn to remove **noise** in stages, often guided by text, but a real implementation can operate in a **latent representation** rather than direct pixels.

The toy values let us inspect an update without pretending to simulate that trained **process**. An attractive final image still needs checks for labels, requested relations and factual claims.

Limit: Not all image generators use diffusion; this toy step is neither a complete algorithm nor an executed model.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Use Study Buddy generated diagrams only with clear **provenance** and checked educational labels. Draw your **input**, deciding step, result and limit.

Supported question: What may these two grids explain, and what can they not establish about a real generator?

Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.